A sharp quantitative tensor-splitting criterion without the zero-growth hypothesis

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Status. Strengthened partial result, likely valid, for human review. It removes the source theorem's zero-growth hypothesis when all but at most one factors have growth-normalized similarity constant no larger than the tensor product's similarity constant. These factors may be nonnormal. The general question remains open.

1 Source question

Let $\mathcal{T} = (T(t))_{t\geq 0}$ be a semigroup on a Hilbert space. Write $e_d\mathcal{T} = (e^{dt}T(t))_{t\geq 0}$, let $\omega_0(\mathcal{T})$ denote its exponential growth bound, and let $\mathcal{C}(\mathcal{T})$ be the optimal condition number of a similarity taking $\mathcal{T}$ to a contraction semigroup. Oliva-Maza and Tomilov prove that if $\bigotimes_{k=1}^m \mathcal{T}_k \in \mathcal{SC}$ and its growth bound is zero, then there are $d_k \in \mathbb{R}$ with $\sum_k d_k = 0$ and

$$\max_k \mathcal{C}(e_{d_k}\mathcal{T}_k) \leq \mathcal{C}\left(\bigotimes_{k=1}^m \mathcal{T}_k\right).$$

Immediately afterwards they ask whether the zero-growth assumption is redundant [1, Thm. 5.3 and p. 18].

In the proof of Theorem 1.1(ii), if $\omega_0(\mathcal{T}_1 \otimes \mathcal{T}_2) = 0$, then for each $n \in \mathbb{N}$ one may define the equivalent inner product $\langle \cdot, \cdot \rangle_n'$ on H_2 as

$$\langle f, g \rangle_n' := \frac{1}{D_n} \int_0^n \langle T_1(t)h_n \otimes f, T_1(t)h_n \otimes g \rangle_{\mathrm{eq}} \, dt, \qquad D_n := \int_0^n \|T_1(t)h_n\|^2 \, dt,$$

for all $f \in H_2$ and $n \in \mathbb{N}$. Then the corresponding norm on H_2 satisfies

$$\|f\| \le \|f\|_n' \le \mathcal{C}(\mathcal{T}_1 \otimes \mathcal{T}_2)\|f\|, \qquad f \in H_2, \, n \in \mathbb{N},$$

hence $\mathcal{C}(\mathcal{T}_2) \le \mathcal{C}(\mathcal{T}_1 \otimes \mathcal{T}_2)$. The argument used in the induction step from the proof above allows to extend the "only if" part of Theorem 1.1(ii) to the following form.

Theorem 5.3. *For each $k \in \mathbb{N}_m$, let H_k be a Hilbert space and let $\mathcal{T}_k = (T_k(t))_{t \ge 0} \subset \mathcal{L}(H_k)$ be a nonzero semigroup strongly continuous in $(0, \infty)$ such that $\otimes_{k=1}^m \mathcal{T}_k \in \mathcal{SC}\left(\otimes_{k=1}^m H_k\right)$ and $\omega_0\left(\otimes_{k=1}^m \mathcal{T}_k\right) = 0$. Then there is $(d_k)_{k=1}^m \subset \mathbb{R}$ with $\sum_{k=1}^m d_k = 0$ such that*

$$\max_{k \in \mathbb{N}_m} \mathcal{C}(e_{d_k}\mathcal{T}_k) \le \mathcal{C}\left(\otimes_{k=1}^m \mathcal{T}_k\right).$$

We do not know if the assumption $\omega_0\left(\otimes_{k=1}^m \mathcal{T}_k\right) = 0$ is redundant in the statement above.

To finish this section, we employ Theorem 1.1(ii) to characterize semigroups $\mathcal{T}$ such that $\mathcal{T} \otimes \mathcal{S}$ is similar to a contraction semigroup if and only if $\mathcal{S}$ is similar to a contraction semigroup. This characterization seems to be of value in applications.

Let H be a Hilbert space, and let $\mathcal{T}$ be a semigroup on H strongly continuous in $(0, \infty)$. We say that $\mathcal{T}$ *tensorially preserves similarity to contraction semigroups* if, for every Hilbert space K and every semigroup $\mathcal{S}$ on K strongly continuous in $(0, \infty)$, $\mathcal{T} \otimes \mathcal{S} \in \mathcal{SC}(H \otimes K)$ is equivalent to $\mathcal{S} \in \mathcal{SC}(K)$.

Proposition 5.4. *Let H be a Hilbert space, and let $\mathcal{T} = (T(t))_{t \ge 0} \subset \mathcal{L}(H)$ be a semigroup strongly continuous in $(0, \infty)$. Then $\mathcal{T}$ tensorially preserves similarity to contraction semi-*

2 Proof intuition

The source paper's complete-boundedness proof contains more quantitative information than its nonzero-growth splitting argument uses. If one groups a factor $\mathcal{T}_1$ against the tensor product of all remaining factors, then normalizing the complementary tensor product to spectral radius one extracts $\mathcal{T}_1$ with no loss in the similarity constant. In general this produces an endpoint shift incompatible with the endpoint shifts extracted for the other factors. The incompatibility disappears whenever the complementary factors already have similarity constants at most the available product constant at their growth-normalized endpoints: their endpoint shifts can simply be used, and their sum supplies exactly the opposite shift for $\mathcal{T}_1$. Contractivity and normality are useful sufficient conditions, but they are not necessary.

3 Strengthened partial theorem

Theorem 1. *Let $m \ge 2$, and for $1 \le k \le m$ let $\mathcal{T}_k$ be a nonzero semigroup on a Hilbert space, strongly continuous on $(0, \infty)$. Assume*

$$\mathcal{T} := \bigotimes_{k=1}^m \mathcal{T}_k \in \mathcal{SC}$$

and put $C = \mathcal{C}(\mathcal{T})$. Suppose that the numbers $\beta_k := \omega_0(\mathcal{T}_k)$ are finite for $2 \le k \le m$ and that

$$\mathcal{C}(e_{-\beta_k}\mathcal{T}_k) \le C \qquad (2 \le k \le m). \tag{1}$$

Define

$$d_1 := \sum_{k=2}^{m} \beta_k, \qquad d_k := -\beta_k \quad (2 \le k \le m).$$

Then $\sum_{k=1}^{m} d_k = 0$ *and*

$$\max_{1\le k\le m} \mathcal{C}(e_{d_k}\mathcal{T}_k) \le C.$$

No assumption on $\omega_0(\mathcal{T})$ *is required.*

Proof. Set

$$\mathcal{S} := \bigotimes_{k=2}^{m} \mathcal{T}_k, \qquad \beta := \omega_0(\mathcal{S}) = \sum_{k=2}^{m} \beta_k.$$

The equality of growth bounds follows from the tensor-norm identity, as in the source paper. Since β is finite, $\mathcal{S}$ is not quasinilpotent.

The endpoint estimate established in the alternative complete-boundedness proof of the finite splitting theorem in [1, Sec. 8], applied to the two factors $\mathcal{T}_1$ and $\mathcal{S}$, gives

$$\mathcal{C}(e_\beta \mathcal{T}_1) \le \mathcal{C}(\mathcal{T}_1 \otimes \mathcal{S}) = C. \tag{2}$$

Indeed, for every $t > 0$ the operator $e^{-\beta t}S(t)$ has spectral radius one. The Paulsen–Pearcy–Petrović endpoint estimate bounds the similarity constant of $e^{\beta t}T_1(t)$ by C; Proposition 8.3 of the source then passes the uniform pointwise estimate to the whole semigroup, yielding (2).

For $k \ge 2$, hypothesis (1) says directly that

$$\mathcal{C}(e_{d_k}\mathcal{T}_k) = \mathcal{C}(e_{-\beta_k}\mathcal{T}_k) \le C.$$

Moreover $d_1 = \beta$, so (2) supplies the remaining factor. Finally,

$$\sum_{k=1}^{m} d_k = \sum_{k=2}^{m} \beta_k - \sum_{k=2}^{m} \beta_k = 0.$$

This proves the claim. □

Corollary 2. *The conclusion of Theorem 1 holds if* $\mathcal{T}_k$ *is a normal semigroup for every* $2 \le k \le m$ *and each* $\omega_0(\mathcal{T}_k)$ *is finite. In particular, for two factors the zero-growth hypothesis is unnecessary whenever either factor is normal with finite growth bound.*

Proof. For a normal operator, norm and spectral radius agree. Therefore, for $k \ge 2$ and every $t > 0$,

$$\left\| e^{-\beta_k t}T_k(t) \right\| = e^{-\beta_k t} r(T_k(t)) = 1.$$

Thus each $e_{-\beta_k}\mathcal{T}_k$ is contractive and has similarity constant one, so (1) holds. Therefore Theorem 1 applies. □

For two factors, Theorem 1 has the following useful form. If $\beta = \omega_0(\mathcal{T}_2)$ is finite and

$$\mathcal{C}(e_{-\beta}\mathcal{T}_2) \le \mathcal{C}(\mathcal{T}_1 \otimes \mathcal{T}_2),$$

then the sharp conclusion holds with shifts $d_1 = \beta$ and $d_2 = -\beta$. This includes nonnormal factors whenever their endpoint similarity can be implemented within the product's distortion budget.

4 Finite-dimensional endpoint equality

For a complex matrix A and $C \geq 1$, define

$$q_C(A) := \inf_{\substack{P>0 \\ \kappa(P)\leq C^2}} \frac{1}{2}\lambda_{\max}\Big(P^{-1/2}(A^*P + PA)P^{-1/2}\Big),$$

and write $s(A) = \max\{\operatorname{Re}\lambda : \lambda \in \sigma(A)\}$.

Theorem 3. *Let $A \in M_n(\mathbb{C})$, $B \in M_r(\mathbb{C})$, $C \geq 1$, and $K = A \otimes I + I \otimes B$. If $q_C(B) = s(B)$, then*

$$q_C(K) \geq q_C(A) + q_C(B).$$

The same conclusion holds with A and B interchanged.

Proof. Fix $\gamma > q_C(K)$. After multiplying a feasible metric by a positive scalar, there is $P > 0$ such that

$$I \leq P \leq C^2 I, \qquad K^*P + PK \leq 2\gamma P.$$

Choose a unit eigenvector v of B with eigenvalue λ satisfying $\operatorname{Re}\lambda = s(B)$, and compress P to

$$P_v = (I \otimes v)^* P (I \otimes v) > 0.$$

Compression preserves the order bounds, so $I \leq P_v \leq C^2 I$. Because $Bv = \lambda v$, compression of the Lyapunov inequality gives

$$A^*P_v + P_v A + 2\operatorname{Re}\lambda\, P_v \leq 2\gamma P_v.$$

Consequently $q_C(A) \leq \gamma - s(B)$. Letting $\gamma \downarrow q_C(K)$ and using $q_C(B) = s(B)$ proves the assertion. □

The equality hypothesis $q_C(B) = s(B)$ says precisely that the growth-normalized matrix semigroup can be made contractive within distortion C. Theorem 3 is therefore the infinitesimal matrix analogue of Theorem 1.

5 Verification, limitations, and the remaining obstruction

The signs in the proof are forced: normalizing the complement uses $e_{-\beta}\mathcal{S}$, while the endpoint extraction gives $e_\beta \mathcal{T}_1$. The individual complementary shifts sum to $-\beta$, so the complete list has zero sum.

The argument does not settle the general negative-growth case. If neither factor attains its spectral abscissa within the product's distortion budget, the endpoint estimates supplied by the source may correspond to incompatible zero-sum shifts. The missing finite-dimensional implication is the constrained-Lyapunov superadditivity inequality

$$q_C(A_1 \otimes I + I \otimes A_2) \geq q_C(A_1) + q_C(A_2).$$

This inequality is not proved here. Convex SDP bisection described in the accompanying attempt note tested more than 2,300 random real pairs in factor dimensions two through four, in addition to structured Jordan families. No strictly active violation was found. Numerical boundary artifacts were discarded using the rigorous endpoint equality in Theorem 3. These computations are evidence only and play no role in either theorem.

A bounded search on 21 July 2026 checked the original arXiv record, the June 2026 sequel on infinite tensor products, and keyword searches for the exact redundancy question. No later resolution of the general finite quantitative question was located. This packet claims only the stated partial theorem and does not claim novelty beyond isolating the corollary and the remaining metric obstruction.

Human-review recommendation. Verify the endpoint passage from Theorem 8.2 and Proposition 8.3 of the source paper and the compression argument in Theorem 3; retain the packet as a strengthened partial result. Further work should focus on the strictly active constrained-Lyapunov regime, where both factor endpoint similarity constants exceed the available distortion budget.

References

[1] J. Oliva-Maza and Y. Tomilov, *On similarity to contraction semigroups and tensor products, I*, J. Funct. Anal. **291** (2026), no. 3, Paper 111492; arXiv:2509.01005.